**EXPERIMENT 14**

# Exploring Pro-Code Extension of a Zoho Creator Application Using Zoho Catalyst and Custom Functions

Patient Appointment Scheduling System — Priority Scoring Extension Use Case

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## 1. Aim

To explore how the Patient Appointment Scheduling System built in Zoho Creator can be extended beyond low-code, using reusable Deluge custom functions and a pro-code serverless function on Zoho Catalyst, so that logic too complex for simple workflow actions can still be added to the application.

The exploration uses an appointment priority scoring example to show how the two parts connect, when each option should be used, and what care is needed in a real implementation.

## 2. Procedure

**Step 1: Identify the extension need** — The existing Patient Profile form and the workflows from Experiments 6, 7 and 10 were reviewed. A new requirement was chosen that built-in actions handle poorly: calculating a priority level for each appointment from the patient's age and existing conditions.

**Step 2: Compare the extension options** — Workflow actions, Deluge custom functions, a Creator Custom API and a Catalyst function were compared on skill needed, power and typical use (Section 3.2).

**Step 3: Design the architecture** — The Creator side (form, workflow, custom function, connection) and the Catalyst side (Advanced I/O function and optional storage) were laid out in an architecture diagram (Section 3.1).

**Step 4: Write the Deluge custom function** — A reusable standalone function was drafted that builds a JSON payload and calls the Catalyst endpoint through a connection using invokeurl (Section 3.3).

**Step 5: Write the Catalyst function** — A Node.js Advanced I/O function was drafted that receives the payload, applies the scoring rules and returns the priority as JSON (Section 3.4).

**Step 6: Plan the execution flow** — The order of events from form submission to record update, including the error path, was drawn as a flowchart (Section 3.5), and a decision guide for choosing the extension type was prepared (Section 3.6).

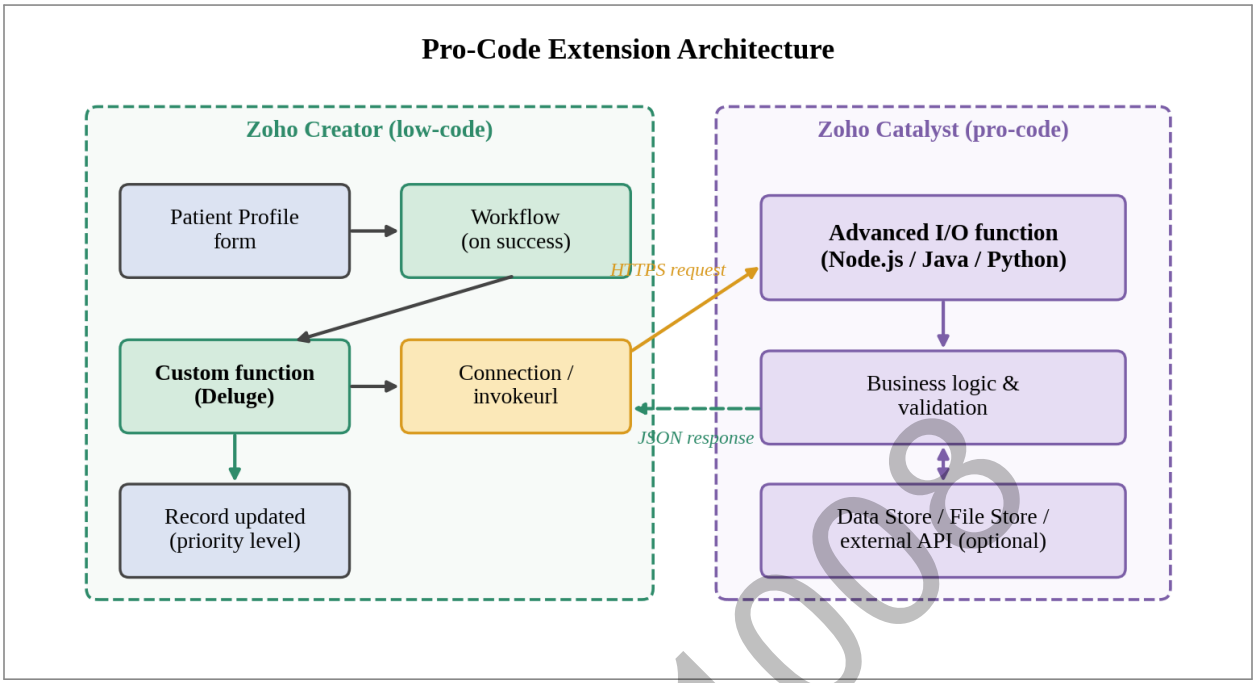
**Step 7: Note the precautions** — Security, error handling, limits and maintenance points for such an integration were listed (Section 3.7).

*Note: this is an exploration of the approach. The code shown is illustrative and was not deployed or executed in this record, and the endpoint and connection names are placeholders.*

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# 3. Output

## 3.1 Architecture of the pro-code extension




Zoho Creator keeps the form and workflow. On a successful submission the workflow calls a Deluge custom function, which sends an HTTPS request through a connection to an Advanced I/O function on Zoho Catalyst. The Catalyst function applies the business logic, optionally uses its own storage or an external API, and returns a JSON response that is used to update the record.

## 3.2 Comparison of extension options

| Option | What it is | Skill needed | Best used for |
|---|---|---|---|
| Workflow actions | Built-in actions such as notification, add or update record. | Low (no code) | Standard, simple automation. |
| Deluge custom function | Reusable standalone function written in Deluge and called from workflows. | Low to medium | Reusable business rules inside Creator. |
| Creator Custom API | Own REST endpoint exposed by the Creator application. | Medium | Letting other systems call into Creator. |
| Catalyst function | Serverless function in Node.js, Java or Python running outside Creator, called over HTTPS. | Medium to high (pro-code) | Complex logic, libraries, heavy processing, outside systems. |

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### 3.3 Deluge custom function (illustrative)

```
// Standalone function: getPriorityScore
string getPriorityScore(int age, string conditions)
{
    payload = Map();
    payload.put("age", age);
    payload.put("conditions", conditions);

    response = invokeurl
    [
        url :"<Catalyst Advanced I/O endpoint>/score"
        type :POST
        parameters :payload.toString()
        headers :{"Content-Type":"application/json"}
        connection :"catalyst_connection"
    ];
    return response.get("priority");
}

// Called from the Patient Profile workflow (on success)
priority = thisapp.getPriorityScore(input.age, input.existing_conditions.toString());
Patient_Profile[ID == input.ID].priority_level = priority;
```

The custom function packages the request once so any workflow can reuse it. The workflow passes the age and conditions, receives the priority and stores it, using the same criteria-based fetch with ID == input.ID as in Experiment 10. The URL and connection name are placeholders.

### 3.4 Catalyst Advanced I/O function in Node.js (illustrative)

```
const express = require("express");
const app = express();
app.use(express.json());

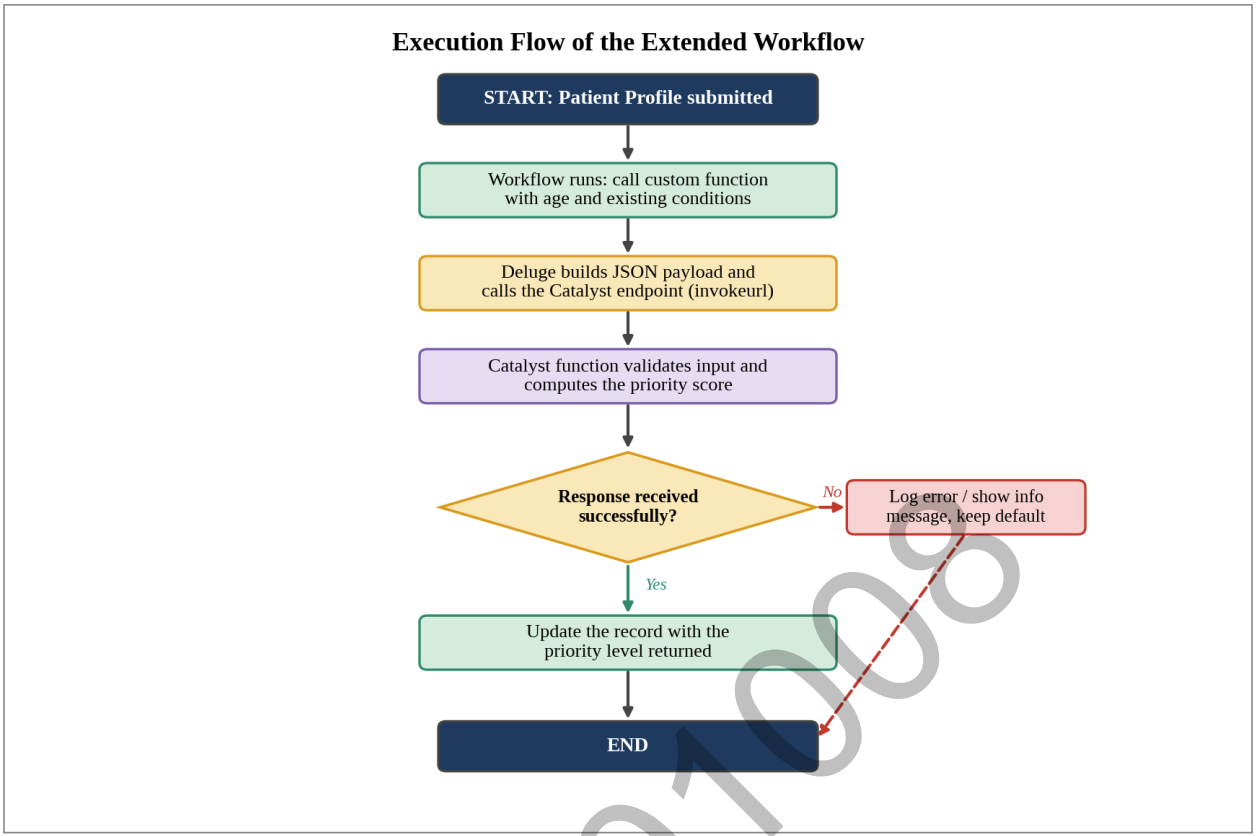
app.post("/score", (req, res) => {
  const { age, conditions } = req.body;
  let score = 0;
  if (age >= 60) score += 2;
  if (conditions && conditions.includes("Heart Disease")) score += 3;
  if (conditions && conditions.includes("Diabetes")) score += 1;
  const priority = score >= 4 ? "High" : score >= 2 ? "Medium" : "Normal";
  res.status(200).json({ priority });
});

module.exports = app;
```

The function reads the JSON body, applies simple scoring rules and returns the priority. The rules are only an example for demonstration; real clinical rules would have to be defined and approved by medical staff.

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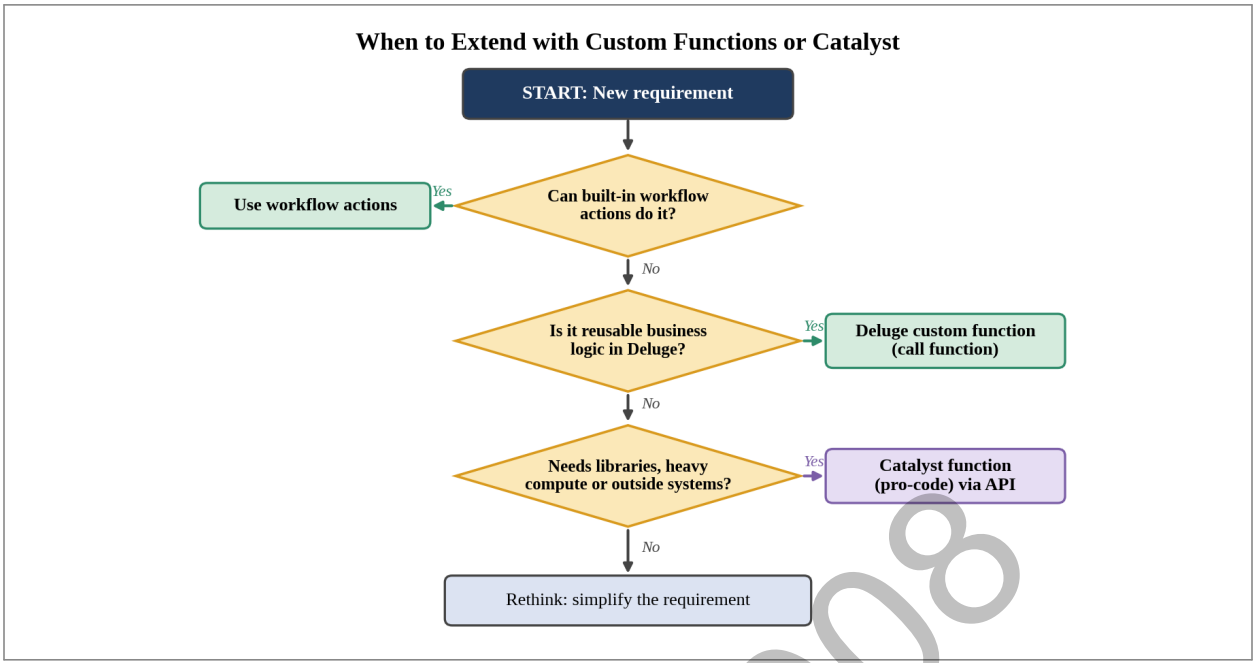
### 3.5 Execution flow of the extended workflow




The flowchart shows the path from form submission to record update. If the response is not received successfully, the error is logged or an informational message is shown and the default priority is kept, so the appointment is never lost because of the extension.

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### 3.6 Decision guide for choosing the extension type




The guide first prefers built-in workflow actions, then a Deluge custom function for reusable Creator logic, and uses a Catalyst function only when libraries, heavy computation or outside systems are needed. If none apply, the requirement should be simplified.

### 3.7 Benefits and precautions

| Aspect | Benefit | Precaution |
|---|---|---|
| Flexibility | Any logic or library can be used in a language such as Node.js, Java or Python. | More code to write, test and maintain than low-code. |
| Reuse | A custom function is written once and called from many workflows. | Keep the function interface (inputs and outputs) stable. |
| Separation | Heavy work runs outside Creator, keeping the application light. | Adds a network call, so add timeouts and error handling. |
| Security | Credentials stay in a connection instead of the script. | Do not put keys in code; restrict access and validate input. |
| Limits and cost | Serverless scaling handled by the platform. | Check current execution limits and pricing in the documentation. |

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## 4. Final Result

The exploration showed how a low-code Zoho Creator application can be extended in two steps: a reusable Deluge custom function inside Creator, and a pro-code Zoho Catalyst function reached over HTTPS through a connection. The architecture, illustrative code, execution flow and decision guide were prepared for a priority scoring extension of the Patient Appointment Scheduling System.

The comparison showed that workflow actions and custom functions are enough for most needs, and that Catalyst should be added only for complex logic, libraries or outside systems, with attention to security, error handling and limits.

The exercise met its objective of exploring pro-code extension. The code was not deployed or tested in this record, so results of a live run are not claimed.